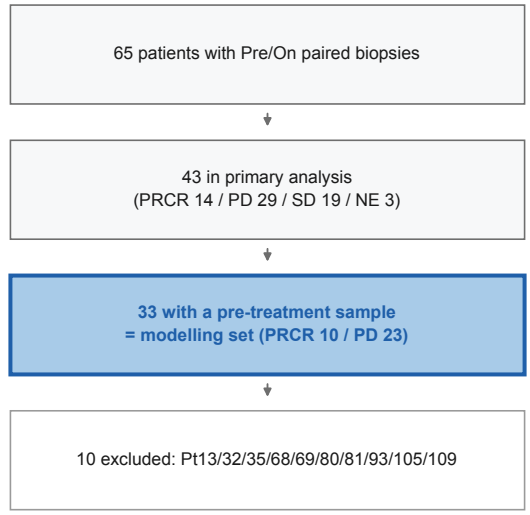
(a) Cohorts and their roles

cohort	smp / pat	role in this study
GSE91061	109 / 65	development
GSE78220	28 / 26	external validation
GSE215868	105 / 105	large-scale validation
GSE244982	41 / 41	direction consistency
GSE308433/4/5	42 / 34	paired tumour–TCR

(b) Analysis unit: patient



(c) Analysis-unit checks performed

situation	handling
Repeated biopsies	paired / mixed structures; not independent samples
Pt27A / Pt27B	one patient, two sites; collapsed by median
Pt16 (GSE78220)	on-treatment only; dropped by flag
Mel-TIL-026 / HM026	same patient in two cohorts; de-duplicated
GSE215868 plate ID	plate position, not a patient ID

(d) What may and may not cross cohorts

1. Sampling time	GSE91061 pairs pre-dose with C1D29; the others are pre-tx
2. Label definitions	verbatim RECIST strings vs. best overall response vs. BOR
3. Platform, units	raw counts / FPKM / featureCounts / NanoString counts

FORBIDDEN

pooling patients or expression matrices across cohorts

PERMITTED

study-level meta-analysis of the per-cohort AUCs computed here, which touches no patient-level record